Applied Econometrics: Wage Determination & Time-Series Autocorrelation

Executive Summary & Empirical Research Overview

Dataset Source: Wooldridge Library (`wage1` & `phillips`)

Methodology: Cross-Sectional OLS & Dynamic Time-Series

EXECUTIVE OVERVIEW

This project conducts an empirical econometric investigation into cross-sectional wage determination using survey data (`wage1`) and evaluates macroeconomic time-series inflation-unemployment dynamics (`phillips`) from the `wooldridge` package. The analysis integrates exploratory data auditing, functional form specification, diagnostic hypothesis testing, robust standard error corrections, and dynamic time-series modeling.

KEY EMPIRICAL FINDINGS

- **Returns to Education & Experience:** Standard Ordinary Least Squares (OLS) log-wage regressions indicate a statistically significant baseline return to schooling of approximately 8.3% per additional year of education, alongside quadratic diminishing marginal returns to overall workforce experience and job tenure.
- **Gender Wage Gap & Interaction Effects:** Incorporating binary indicators reveals a persistent conditional wage gap favoring male workers. Interaction terms (`educ × female`) demonstrate differential returns to education across gender groups, supported by joint *F*-tests for parameter equality.
- **Model Diagnostics & Heteroskedasticity:** Breusch–Pagan and White tests reject the null hypothesis of homoskedastic errors in baseline wage equations. Applying HC1 heteroskedasticity-consistent robust standard errors corrects hypothesis testing and confidence intervals without altering OLS point estimates. Jarque–Bera testing highlights moderate right-skewness in log-wage residuals.
- **Time-Series Autocorrelation & Specification:** Diagnostic testing on the expectations-augmented Phillips curve reveals significant positive *AR(1)* serial correlation via Durbin–Watson and Breusch–Godfrey tests, demonstrating the necessity of dynamic specification and explicit lag structures in macroeconomic models.

ECONOMETRIC METHODOLOGY SUMMARY

Domain	Econometric Model	Diagnostic Tests	Robustness / Corrections
Cross-Sectional (`wage1`)	Semi-log OLS, Interactions	Breusch-Pagan, White, Jarque-Bera	HC1 Heteroskedasticity-Consistent SEs
Time-Series (`phillips`)	Phillips Curve Specification	Durbin-Watson, Breusch-Godfrey	Ramsey RESET, First-Differencing / Lags

> **Replicability Note:** All empirical estimations, diagnostic matrices, and figures are fully programmatic and executable directly via the integrated Google Colab Jupyter Notebook in the repository repository.